

ChannelFormer: a *Transformer-based* model for fNIRS missing-channel reconstruction

From quality control in paediatric ASD brain-imaging data to an idea for EEG bad-channel reconstruction

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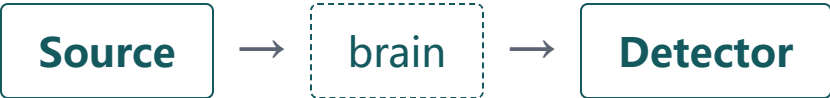
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Why this paper: aggressive bad-channel rejection in our ASD EEG recordings leaves too many interpolated channels. This paper asks the same question in fNIRS — **can a lost channel be rebuilt from the channels that survived?**

Why is *whole-channel loss* so troublesome?

fNIRS in one line



One source–detector pair = one channel.
Near-infrared light → ΔHbO / ΔHbR → cortical hemodynamics.

Children make this worse

- Head movement & probe displacement
- Poor scalp contact, motion artifact
- ASD children: less compliance, more movement — the paper’s core clinical motivation

The paper does not study a few dirty seconds inside one channel. It targets an entire channel whose time series is missing.

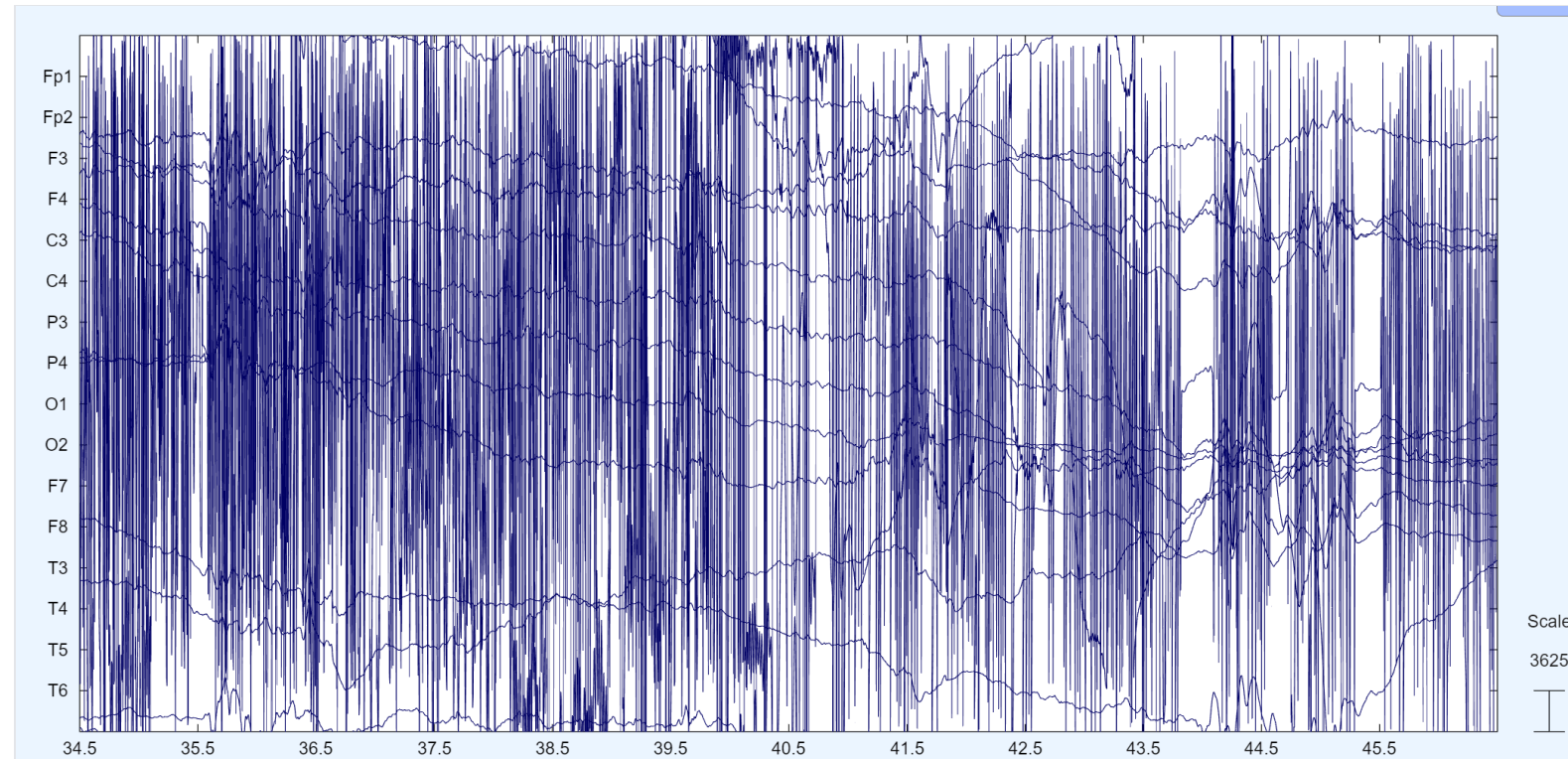
healthy recording		after probe displacement	
CH1	~~~~~	CH1	~~~~~
CH2	~~~~~	CH2	XXXXXXX ← gone
CH3	~~~~~	CH3	~~~~~
CH4	~~~~~	CH4	~~~~~

FIG. 1 — PROBE DISPLACEMENT KILLS THE WHOLE SERIES

Downstream **connectivity, brain-network and classification analyses** all need as complete a multi-channel matrix as possible — a lost variable cannot be filtered away like a spike.

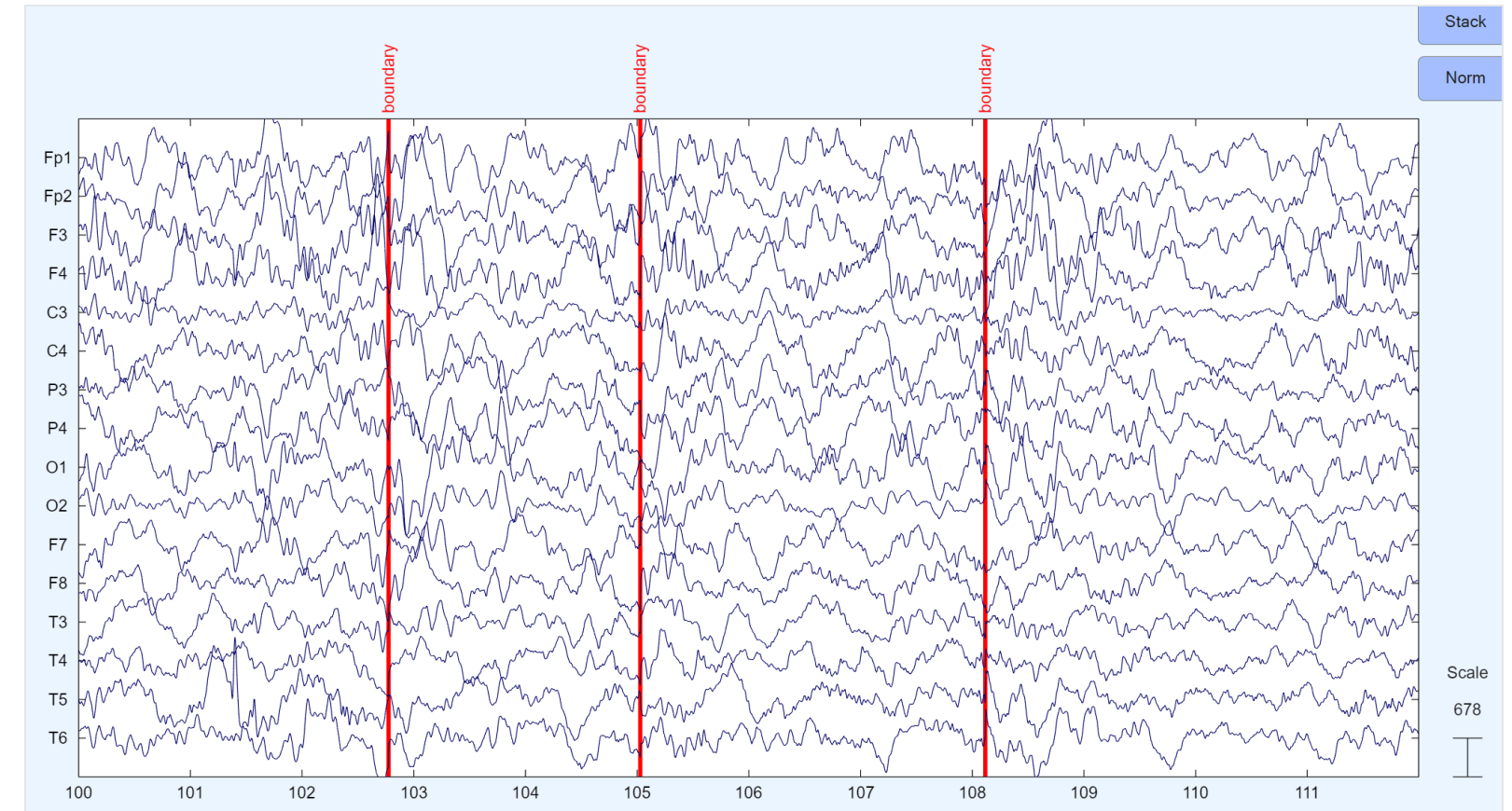
This is exactly what our *own recordings* look like

FIG. A — RAW RECORDING, ASD CHILD



Before any cleaning: dense high-amplitude artifacts swamp the record (Scale 3625). Several channels carry almost no usable brain signal — classic bad-channel territory.

FIG. B — AFTER REJECTION + SPLINE INTERPOLATION

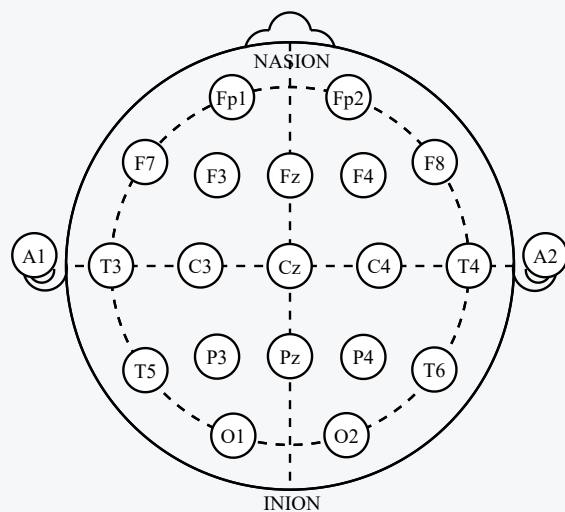


After traditional handling: rejected segments leave red **boundaries**; whole stretches of data are lost, and interpolated channels are severe distortions of the true signal — too many boundaries fragment downstream analysis.

The pain point: aggressive bad-channel rejection → massive interpolation → boundaries everywhere. Better channel reconstruction is not a luxury — it is what our pipeline is missing.

How *spherical spline interpolation* fills a bad channel

The setting



International 10-20 system (Wikimedia) — our case: **F3 bad**, the rest usable

Core assumption: the scalp potential field $V(\mathbf{r}, t)$ is a **spatially smooth function** on the scalp sphere — F3 is not unrelated to its neighbours.

Not a neighbour average

$$F_3 \neq \frac{1}{4} (Fp1 + F7 + Fz + C3)$$

And not inverse-distance weighting either — a smooth field is fitted through **all** electrodes' spatial relations on the sphere.

Repeated at every time point

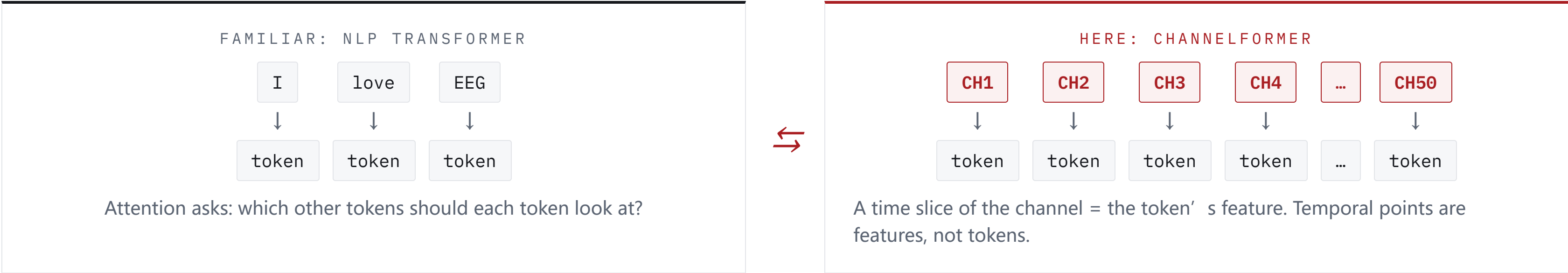
```
t = 1 ms  all good electrodes → solve  $c_i(t)$  →  $\hat{F}3$ 
t = 2 ms  repeat →  $\hat{F}3$ 
...
↓
whole series  $\hat{F}3(t)$ 
```

$$\hat{V}(\mathbf{r}_0, t) = c_0(t) + \sum_{i=1}^N c_i(t) g(\cos \theta_{i0})$$

$\mathbf{r}_0$ position of the bad electrode · i the remaining good electrodes · θ_{i0} spherical angle between electrode i and the bad site · $g(\cdot)$ spherical spline kernel · $c_i(t)$ coefficients solved from all good electrodes at each time point

What it actually uses: **electrode geometry + the instantaneous spatial potential field**. No training, no learned $P(F_3 \mid F_7, F_z, C_3, \dots)$ from large data — it's **model-based spatial interpolation**, not data-driven reconstruction.

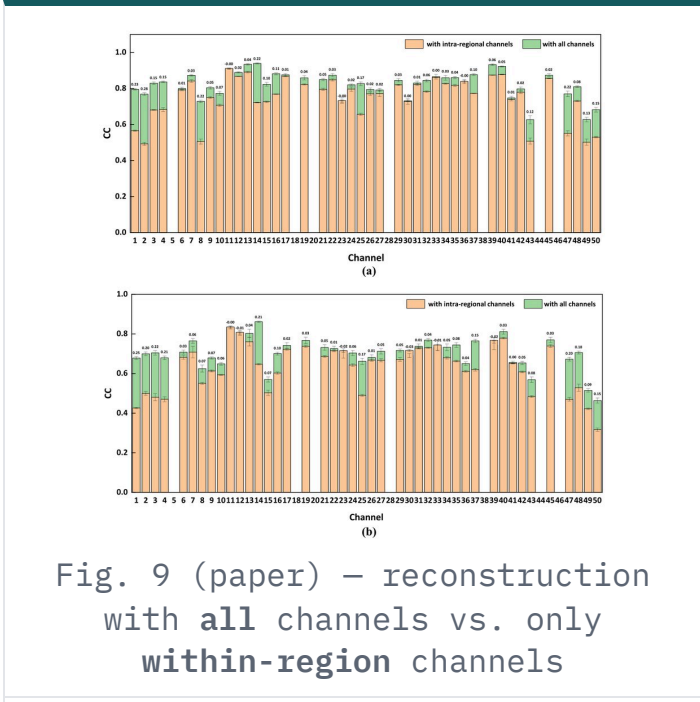
Treat *channels as tokens*



1 channel = 1 token

its time series = **token feature**

With $C = 50$ channels and slice length T , the input is $X \in \mathbb{R}^{50 \times T}$ — fundamentally different from time-step-as-token Transformers.



- Why this reframing is the whole point
- What is lost is an **entire variable**, not a few noisy points — a dead channel removes a whole column of the data matrix.
 - The 50 channels are **not independent**: the scalp field is smooth, so neighbours constrain the gap — exactly what attention can exploit.
 - Time goes into **features, not tokens**: tokens stay few but long, matching fNIRS' s slow dynamics.

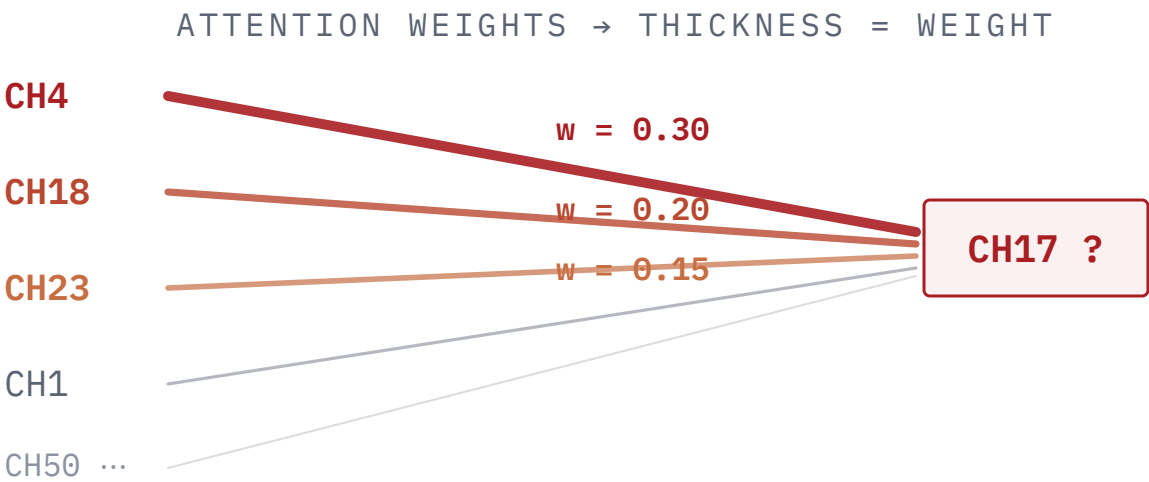
What does *self-attention* actually do here?

THE GENERAL RULE

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

No need to memorise it — it computes **weighted relationships between every pair of tokens**. Here, the tokens are channels.

APPLIED TO ONE LOST CHANNEL – SAME RELATIONSHIP, DRAWN



$$\widehat{CH}_{17} \approx 0.30 CH_4 + 0.20 CH_{18} + 0.15 CH_{23} + \dots$$

The weights in the equation **are** the labelled arrows above — one relationship, two views. (Not literally a linear blend; it is the intuition that holds.)

Multi-head = different relationship types

- Head 1 → nearby channels
- Head 2 → global co-fluctuation
- Head 3 → other dependencies

⚠ attention weight ≠ functional connectivity

Model attention is a computational mixture — it must **not** be read as a neurophysiological connection.

Three key designs, in *plain words*

① Binary mask — “tell the model what is real”

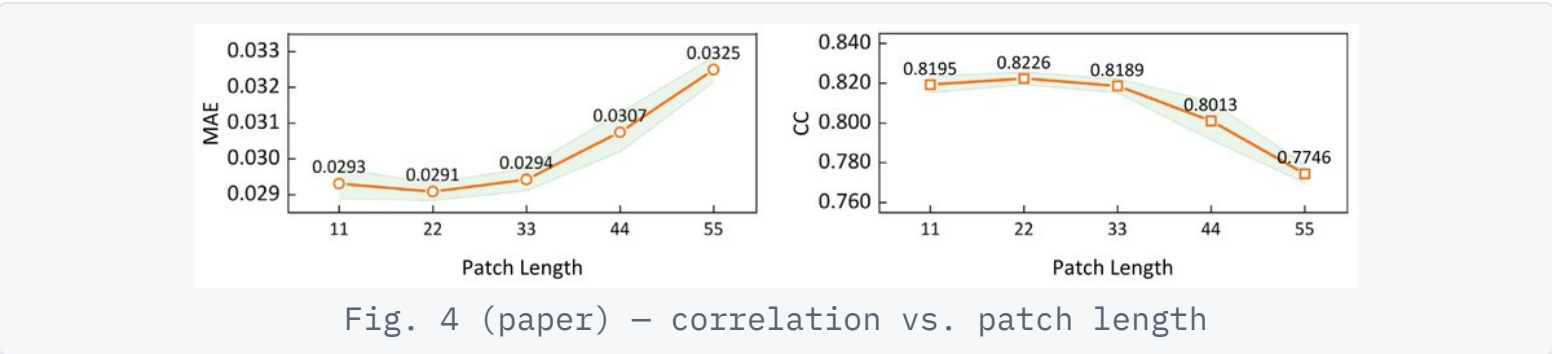
Every channel gets a tag: **1 = really measured, 0 = missing, please invent it.** The model never has to guess which channels it should trust.

1 bad ch.	1 1 1 1 1 0 1 1 1 1 1
several	1 1 0 1 0 1 0 1 1 0 0 1
1 source	1 1 1 1 0 0 1 1 0 0 1 1 1

Third row is the realistic one: one light source feeds several channels — if its scalp contact fails, **CH5, CH6, CH10, CH11 vanish together**, not at random.

② Time slicing — “cut long records into short windows”

A 2-min series is far too long to feed in as one feature. So each channel’ s record is cut into short windows; **each window becomes one token’ s feature.**



- shorter features
- less overfitting
- more training windows

fNIRS signals barely carry long-range temporal dependencies, so little information is lost by cutting.

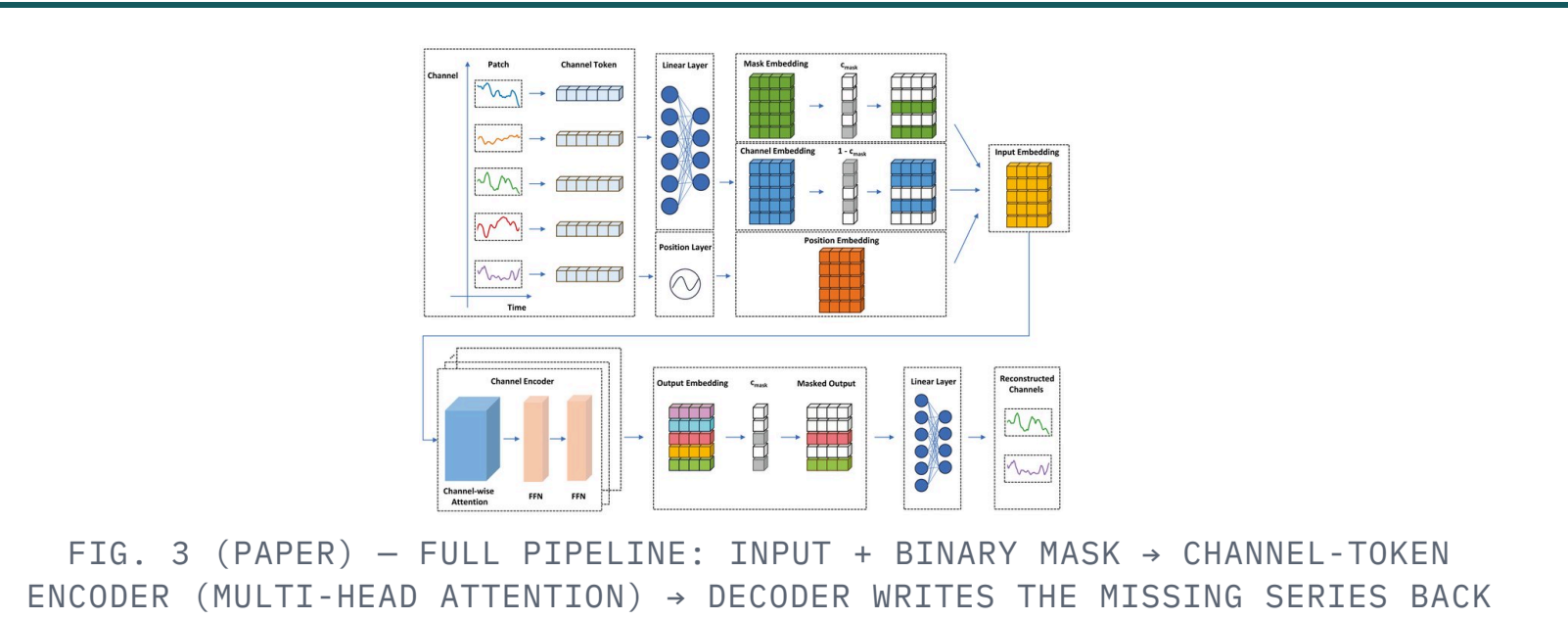
③ Hybrid loss — “values right *and* shape right”

MAE alone can be gamed by a flat average: numbers close, dynamics gone. So the reconstruction is graded on **two things at once**:

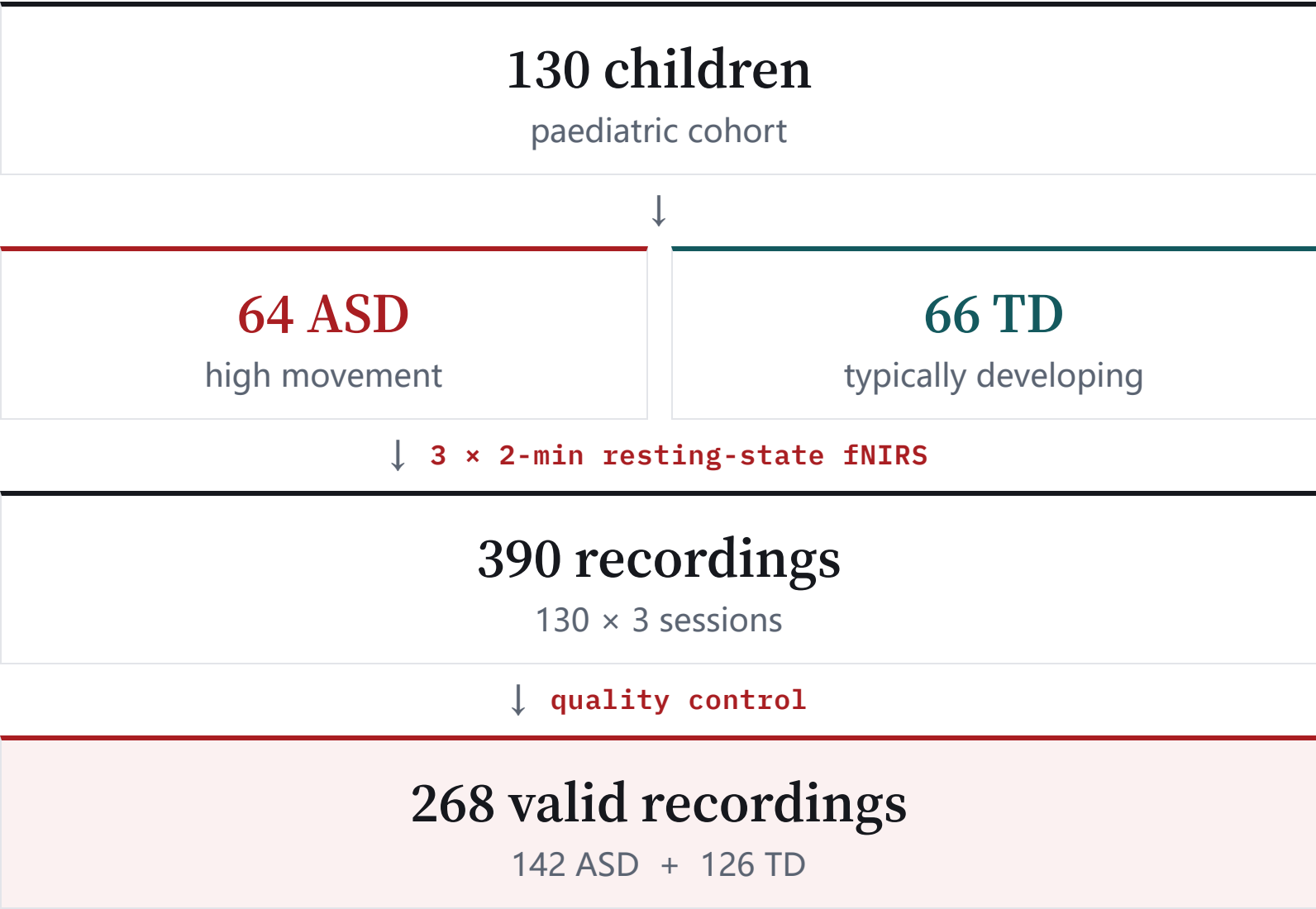
$$L \sim L_{MAE} + \lambda(1 - r)$$

MAE
are the values close to the original?

r
does the reconstructed wave rise and fall in the same places?



It actually uses *ASD children's data*



SYSTEM HuiChuang NirSmart	SAMPLING 11 Hz
WAVELENGTHS 730 / 850 nm	PROBE LAYOUT 23 sources × 16 detectors
CHANNELS 50	

Positioning matters: ChannelFormer is a **signal-reconstruction model** — not an ASD classifier, not a diagnostic model. ASD’ s value here: a high-movement, heterogeneous clinical population as the validation scenario.

Read 268 correctly: 268 recording samples, not 268 children.

How well does it recover *12 missing channels*?

CORRELATION COEFFICIENT	TD	ASD
Single-channel loss	0.8226 ± 0.0032	0.7024 ± 0.0025
12-channel loss	0.8022 ± 0.0018	0.6719 ± 0.0026
Light-source loss	0.7953 ± 0.0804	0.6689 ± 0.0849

Source: ChannelFormer, Applied Soft Computing 2026

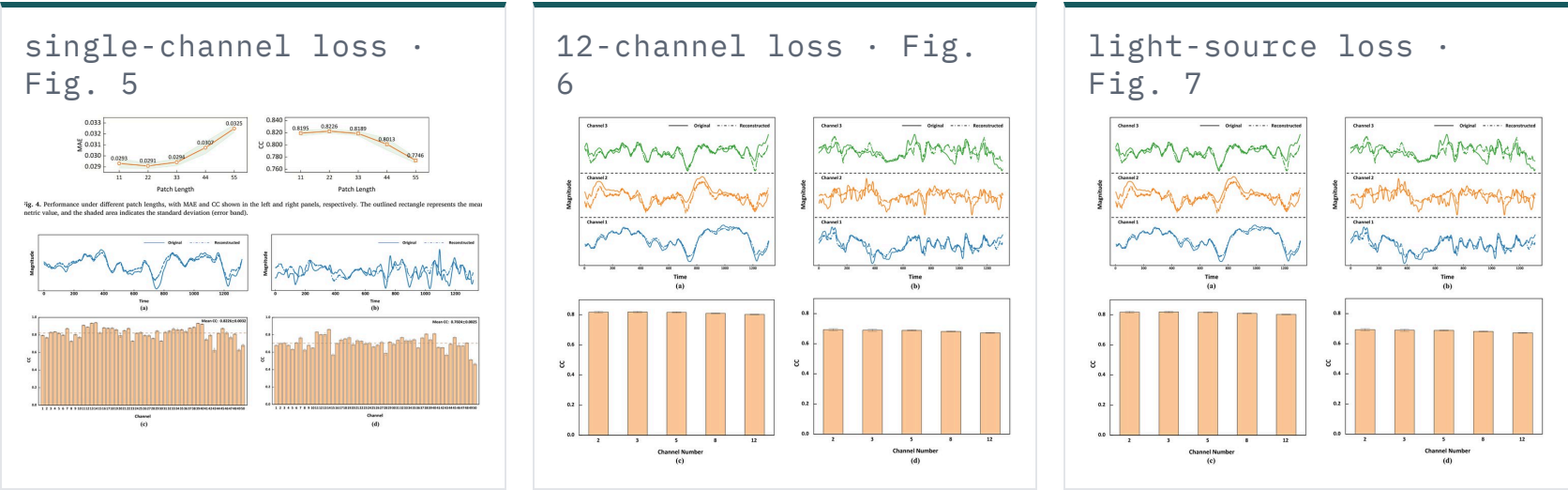


FIG. 5–7 (PAPER) – ORIGINAL VS. RECONSTRUCTED ΔHBO, ONE SCENARIO PER TABLE ROW

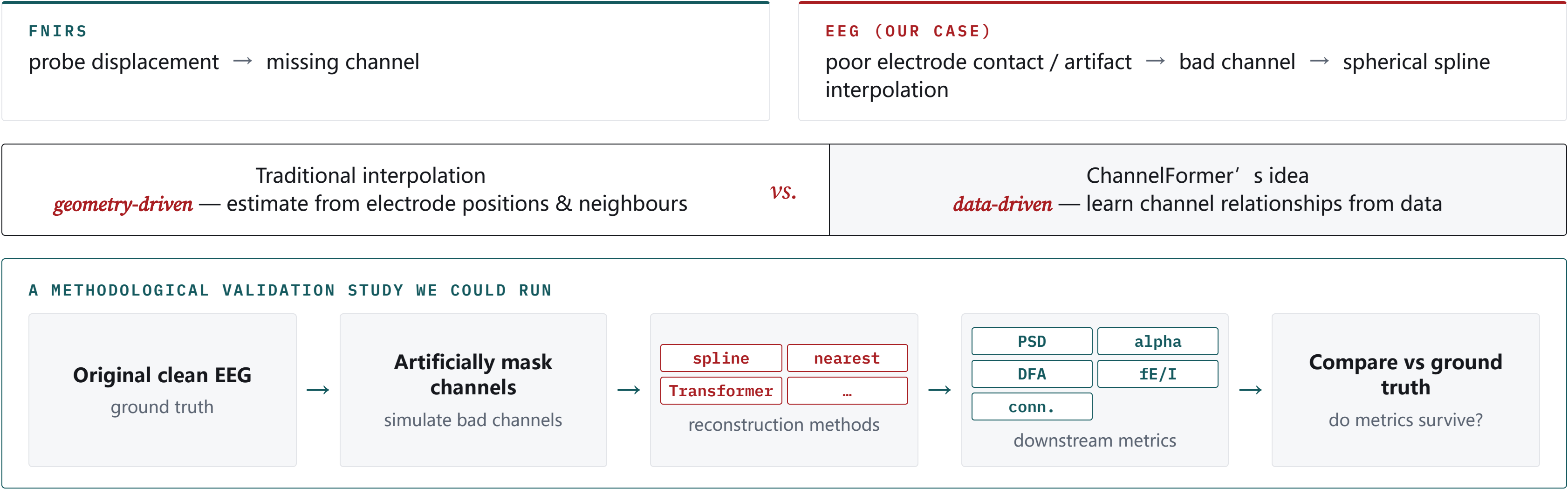
Phenomenon 1. Losing 24% of channels doesn’t collapse performance

TD 0.823 → 0.802 , ASD 0.702 → 0.672 : even with 12/50 channels gone, waveform correlation stays moderately high. fNIRS channels carry **substantial shared information** — cross-channel reconstruction is viable.

Phenomenon 2. ASD is consistently below TD

TD $r \approx 0.80$ vs ASD $r \approx 0.67\text{--}0.70$. Candidate causes: (1) more movement / lower SNR; (2) more heterogeneous neural signals. But **this result alone cannot decide which** — AI does not remove the raw data-quality problem.

What does this mean for *our ASD-EEG* pipeline?



Is ChannelFormer simply better than spline? *Not so fast.*

1 · Needs no training data at all

`pop_interp(EEG, chan, 'spherical')` works immediately — no 1000 subjects, no train/val/test split, no GPU. ChannelFormer first needs **large amounts of complete, high-quality data as ground truth**, artificially masked so the model can learn $X_{\text{masked}} \rightarrow X_{\text{complete}}$.

2 · Generalizes naturally

It depends on **electrode coordinates**, not a training distribution — ASD, TDC, ADHD, adults, children: any reasonable montage works. ChannelFormer risks **domain shift**: trained on awake TDC, tested on chloral-hydrate-sleep ASD means $P_{\text{train}}(\text{EEG}) \neq P_{\text{test}}(\text{EEG})$ — a real warning for our data.

3 · Injects no cohort prior

An ASD-specific ChannelFormer could bake **training-cohort priors** into $\hat{X}_{\text{bad}}$; asking “ASD vs TDC?” on that reconstructed data then becomes **circular**. Spline learns no labels and injects no population prior — it does not manufacture an ASD prior.

	OUR EEGLAB TODAY	CHANNELFORMER
Method	Spherical spline	Transformer reconstruction
Core basis	Electrode spatial geometry	Learned cross-channel relations
Training	Not needed	Needed
Time structure	Moment-wise spatial field	Time series as feature
Non-local relations	Constrained by spatial model	Multi-head attention, global
Many bad channels	Harder as count grows	Trained multi-channel loss
Dataset dependence	Very low	High
ASD generalization	No training needed	Must verify domain shift
Interpretability	Higher	Lower
Compute cost	Extremely low	High
Waveform reconstruction	Robust traditional	Potentially more accurate
Biomarker preservation	Not guaranteed	Also not guaranteed
Connectivity distortion	Present	Possibly more complex
Usable on our EEG now	✓ Yes	✗ Not directly yet

Three things to remember

- 1 ChannelFormer turns **complete channel loss** into a cross-channel reconstruction problem:
 $CH_i \sim P(CH_i \mid CH_1, \dots, CH_{i-1}, CH_{i+1}, \dots)$.
- 2 **Channel-as-token + self-attention** learns global cross-channel dependencies — performance holds even with 24% of channels missing.
- 3 For our ASD-EEG work: **bad-channel handling itself deserves systematic validation** — the reconstruction method may shape the biomarkers we report.

*Can learned EEG reconstruction **preserve physiological biomarkers**?*